

Jiayi Li

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Education

Brown University - Providence, RI	Exp: May 2026
<i>Master of Science in Computer Science</i>	GPA: 3.75
• Coursework: Interactive Computer Graphics, Computational Design and Fabrication	
Boston University - Boston, MA	Jan 2024
<i>Bachelor of Arts in Computer Science and Statistics</i>	GPA: 3.78

Research Experience

Research Assistant , Brown University (PI: David Laidlaw) – Providence, RI	Jun 2025 – Present
• Built an automated data generation pipeline by integrating 4+ AI models (ChatGPT, Flux, SAM, background-removal) in Python, enabling scalable image and text processing for infographic datasets.	
• Curated a high-quality training dataset by evaluating images image-by-image using defined quality criteria, reducing the pool from 37,000+ raw samples to 11,500+ reliable entries and, improving model consistency.	
• Led monthly research syncs and collaborated on dataset design, creating structured infographic variations through text-to-image prompting strategies and reusable prompt templates.	
• Designed an evaluation framework for infographic generation, defining metrics for visual coherence, structural accuracy, and semantic clarity, and benchmarked Style-Aligned to identify performance gaps and guide future comparisons.	

Professional Experience

UI/UX Design Intern , PCCW Global – Beijing, China	Mar 2024 – Jul 2024
• Led user and competitive research using task-flow mapping and feature benchmarking to uncover key usability gaps, directly shaping the direction of two new internal web product features.	
• Proposed 10+ design strategies through interaction analysis and UI trade-off evaluation, improving visual clarity, navigation efficiency, and workflow consistency across the platform.	
• Designed and prototyped responsive interface layouts in close collaboration with PMs, engineers, and UX designers, ensuring design–development alignment and enabling a smoother product handoff process.	

Projects

Nature Werks: Real-Time Natural Environment Renderer , Brown University	Oct 2024 – Dec 2024
• Developed a real-time natural environment renderer in C++/OpenGL (Qt Creator), generating terrain and simulating dynamic weather, sky, and water to support realistic 3D world-building for game environments.	
• Implemented a modular rasterization and lighting pipeline using VAOs/VBOs, FBOs, and custom GLSL shaders, enabling multi-light rendering and interactive scene transitions across different seasonal conditions.	
• youtube.com/NatureWerk .	
Raft Consensus Implementation (Go, Distributed Systems) , Boston University	Jan 2023 – Apr 2023
• Improved the stability of a single-browser request system by implementing the Raft consensus protocol in Go, enabling reliable leader election, log replication, and fault-tolerant coordination across distributed nodes to form a fully functional replicated state machine.	
• Analyzed design trade-offs in a distributed request–response system by running failure-injection tests and timing benchmarks, improving overall system stability and reproducibility.	

Skills

- **Languages:** C++, C, Java, Python, SQL, JavaScript, Go, R, English (fluent), Mandarin (native)
- **Technologies:** Qt Creator, OpenGL, VSCode, Figma, SQLite, Xcode, RStudio